THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT
ANNUAL JOINT EXAMINATION

FORM ONE
031 PHYSICS
MARKING SCHEME NOVEMBER 2024

SECTION A (15 Marks)

1.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
A	A	C	D	C	D	D	D	A	A

01@ = 10 marks

2.

List A:	(i)	(ii)	(iii)	(iv)	(v)
List B:	D	A	E	C	F

01@ = 05 marks

SECTION B (70 Marks)

3. (a) Derived unit: units of measurement derived from the seven SI basic units. 01 mark
Examples: Square mtre (m^2), Cubic metres (cm^3), metres per second (m/s)

(b) Importance of forming hypothesis before doing an experiment: 0½@ = 01½ marks
A prediction about what could happen in an experiment, based on prior knowledge.
01½ marks

(c)
Importance of learning physics in our daily lives

(i)Physics develops fundamental knowledge for future technological advances that will continue to drive the economic engine of the world.
(ii) Physics enhances our understanding of other disciplines such as agriculture, environmental sciences, …

(iii)Physics is an interesting intellectual that inspire young people and expand the frontier of our knowledge about nature…

(iv) Physics helps him to help others; medicine without physics technology would not be so useful. It is also important in acquiring well-paying careers. 01½ @ = 06 marks

4. (a) **First aid is** an immediate care given to an injured person before she/he is taken to a nearby hospital for further medical treatment while first aid kit is a small box containing items that are used to give help to a sick or injured person **and** Usually labeled as "**FIRST AID**" and stored in a safe and easily accessible place (2 marks)

(b)

ITEMS	FUNCTION
i A pair of scissors	To cut adhesive tapes, bandages and gauze
ii 2. Rolls of adhesive tape	To hold firmly into wounds, bandages, gauze and cotton wool.
iii 3. Bandages and cotton wool	To clean and cover wounds.
iv 4.Sterilized new razor blade	Used in treating fresh or old wounds

Any four 02 mark@ = total 08 mark

5. (a) Mass is the quantity of matter in an object while weight is the force of gravity on the object.

02 marks

(b) Seypalm stands on the scale and it reads 54 kg.

(i) Solution:
Data: mass 54 kg
Mass of Juma is 54 Kg 01 mark

(ii) Weight = mass × gravitational force
54 Kg × 10N/Kg
∴Weight = 540N 02 marks

(c) Data: force of gravity on that planet is 2 N/Kg 01 marks
Mass of Seypalm is 54 Kg
Weight = M × g 01 marks
Weight = 54 Kg × 2 N/Kg 01 marks
Weight = 108 N 01 marks
∴ His mass will be 54 Kg and his weight will be 108 N. 01 marks

6. (a) Application of Archimedes' principle
Archimedes principle is employed in determination of relative density of solids heavier or lighter than water. 1 mark

(b) Solution:

Volume of cylindrical solid = $a \times h$

$0.0004\ m^2 \times 0.30$ m = 0.00012 m^3 0½ mark

Mass of cylindrical solid = volume × density

0½ mark

$= 0.00012\ m^3 \times 15000\ Kgm^{-3}$

$= 0.18\ Kg$ 0½ mark

(i) Weight of solid $= m \times g = 0.18\ Kg \times 10\ N/Kg$ 0½ mark
Weight = 1.8 N 0½ mark

(ii) Volume of displaced water = volume of the cylindrical solid 0½ mark
$= 0.00012\ m^3$
Mass of water displaced = Volume of water × density of water
$= 0.00012\ m^3 \times 1000\ Kg/m^{-3}$ 0½ mark
$= 0.12\ Kg$ 0½ mark
Upthrust = mg = 0.12 Kg × 10 N/Kg 0½ mark
∴ Upthrust = 1.2 N 0½ mark

(iii)Apparent weight of the cylinder in water and alcohol.

$= 1.7\ N - 1.2\ N$

$= 0.6\ N$ 0½ mark

Apparent weight of the cylinder in water = 0.6 N

Upthrust due to alcohol = volume × density × g 01 marks

$= 0.00012\ m^3 \times 780\ Kg/m^{-3} \times 10\ ms^{-2}$ 01 mark

= 0.936 N
∴ Upthrust due to alcohol 0.936 N 0½ marks

7. (a) The addition of soap solution reduces the surface tension of water. This reduces the strength of skin that was preventing the pin from sinking, and therefore the pin sinks. 02 mark

(b) Solutin:
(i) Force constant.
Data: Extension (e) = 10 cm = 0.1 m
Force = 0.5 N 0½ mark

Formula:
Force cintant (K) $= \frac{Force}{extension} = \frac{F}{e}$ 01 mark
$K = \frac{F}{e} = \frac{0.5N}{0.1} = 5\ N/m$ 02 marks
∴ The force constant = 5 N/m 01 mark

(ii) Solution
From Hook's law, Force α e; e =20 cm = 0.2 m 01 mark
F = 5 N/m × 0.2 m
F = 1 N 01 mark 01 mark
∴ The required force is 1N. 0½ mark

8. (a)

(i) E-mail 0½ mark

(ii) Shor text message (SMS) 0½ mark

(iii) Fax 0½ mark

(iv) Write a letter 0½ mark

(b)

(i) You should not carry out an experiment if there is no teacher or technician in the laboratory. 01½ mark

(ii) Wash your hands using soap and clean water, put on gloves, wash the cut using salt water and a clean cloth, and finally cover the cuts with adhesives bandages or plaster.

(iii) Burn waste papers away from buildings 01½ mark

(iv) Avoid tasting, eating or drinking anything in the laboratory 01½ mark 01½ mark

(v) Avoid handling apparatus or chemicals in the laboratory unless you are asked to do so by the teacher. Also, experiments should be carried out in the laboratory 01½ mark

Neatness 0½ mark

9. (a)

- Liquid exert pressure when the liquid molecules collides themselves and also when collides with the wall of the containing vessel
- The molecules impart momentum to the walls, producing a force perpendicular to the wall. The sum of the forces of all the molecules striking the wall divided by the area of the wall is defined to be liquid pressure **05 marks**

(b) Data given

Density of water $\rho_w (= 1000 kgm^{-3})$

$l = 135\ mm$ and

$d = 12.5\ mm.$

$density\ of\ oil = ?$

$from$

$presure\ exerted\ on\ oil = presure\ exerted\ on\ water$ …… 01 marks

$\rho_w gl = \rho_o gd$ …… 01 marks

$1000 \times 10 \times 135 = \rho_o \times 10 \times 147.5$ …… 01 marks

$\rho_o = \frac{1000 \times 135}{147.5}$

$\rho_0 = 915\ kg/m^3$ …… 02 marks

SECTION C (15 Marks)

10.

(a)

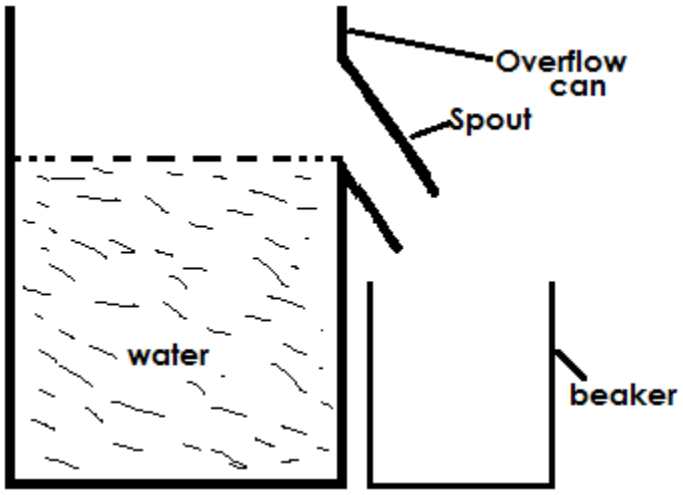
Items	Accuracy(mm)
i. metre rule	**1**
ii. Vernier calliper	**0.1**
iii. Micrometer screw gauge	**0.01**

01 mark@ = total 03 mark

(b) Consider the following steps:-

i Use beam balance measure the mass of stone as M

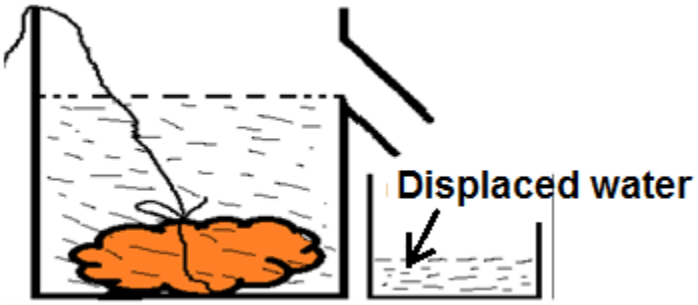
ii Fill the overflow can with water up to the level of the spout



iii Tie the irregular solid (stone) with a string

iv Gently drop the irregular solid into water using a string

v The irregular solid (stone) will displace some water which will be collected in the beaker



vi Transfer the displaced water into a graduated cylinder

vii Measure the volume of water, which is the volume of irregular solid as V

viii Determine the density of stone = $\frac{mass\ of\ stone}{volume\ of\ stone}$

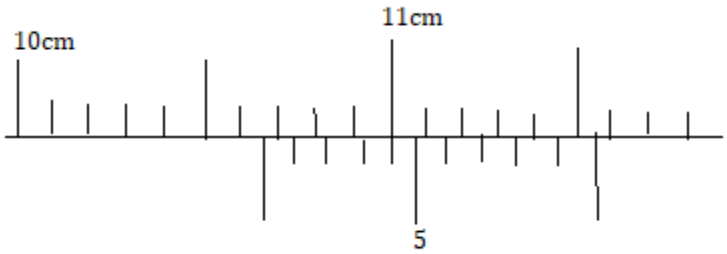
Conclusion

Density of stone = $\frac{M}{V}$

08-mark total

(c) main scale = 10.6cm

Vernier scale = 0.04cm



(4 marks)